

# Code-Layout, Readability and Reusability

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Date          | 05-10-2026                                                           |
| Team ID       | 6ab2256ffc67a5bc022256a7                                             |
| Project Name  | Script_ Controlled ACL _ Restrict Record Access Based on Field Value |
| Maximum Marks | 5 Marks                                                              |

## Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                                                                              |
|------|---------------------------|-----------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | JavaScript code in ACL scripts uses 2-space indentation.                             |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | ACL scripts, Business Rules, and Script Includes are structured cleanly.             |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Variable names like currentRecordState and isAuthorisedUser clearly describe intent. |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions named evaluateRecordAccess() and checkUserRole().                          |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes                           | Key logic for restricting record access based on field value is commented.           |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Reusable ACL logic extracted into a Script Include.                                  |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | Redundant role checks consolidated into a single function.                           |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           | Null checks and exception blocks implemented for invalid record state fields.        |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology          | Where Reused                                       | Reusability Level (High / Medium / Low) |
|------|-------------------------|--------------------------------|----------------------------------------------------|-----------------------------------------|
| 1    | RecordAccessEvaluator   | ServiceNow JS (Script Include) | Multiple ACL rules across incident and task tables | High                                    |
| 2    | FieldRestrictionHelper  | ServiceNow JS                  | Form Client Scripts and UI Policies                | Medium                                  |
| 3    | RolePermissionChecker   | ServiceNow JS                  | ACL Script Conditions and Business Rules           | High                                    |
| 4    | CustomLoggingUtil       | ServiceNow JS                  | System Diagnostics & ACL Audit Logs                | Medium                                  |
| 5    | FieldValueValidator     | ServiceNow JS                  | Data Validation & Pre-execution ACL Checks         | High                                    |

### Overall Code Quality Assessment:

| Aspect                  | Rating (1-5) | Comments                                                                |
|-------------------------|--------------|-------------------------------------------------------------------------|
| Code Layout & Structure | 5            | Clean file hierarchy with well-organized Script Includes and ACL rules. |

|                          |         |                                                                                              |
|--------------------------|---------|----------------------------------------------------------------------------------------------|
| Readability              | 5       | Descriptive variable names, proper indentation, and consistent formatting throughout.        |
| Reusability              | 4       | Core security validation logic extracted into reusable Script Includes.                      |
| Documentation / Comments | 4       | Comprehensive block comments explaining complex field-based condition logic.                 |
| Overall Score            | 4.5 / 5 | Excellent code quality adhering strictly to ServiceNow script guidelines and best practices. |